

PHRAE, Thailand

WMO: 483300

Lat: 18.1286N

Lon: 100.1625E

Elev: 534

StdP: 14.41

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.8	58.5	50.4	55.2	69.4	53.3	61.6	69.9	9.7	85.1	8.5	84.7	0.1	10	0.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	21.8	101.8	77.3	99.3	77.1	97.2	77.0	82.2	92.1	81.4	91.2	80.7	90.3	4.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	157.8	86.8	79.0	153.7	85.7	78.4	150.6	85.0	46.6	92.6	45.7	91.2	44.9	90.6	92.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.3	7.8	DB	51.5	104.4	3.6	2.5	48.9	106.2	46.8	107.6	44.8	109.0	42.2	110.8	(4)
			WB	50.4	83.9	3.4	2.3	47.9	85.6	46.0	86.9	44.1	88.2	41.6	89.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	74.5	78.3	83.4	87.5	85.8	84.5	83.1	82.5	82.8	82.0	78.7	74.6
	DBStd	5.09	3.65	3.45	3.72	3.62	3.84	2.52	2.43	2.08	1.92	2.44	3.58	4.36
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	2	0	0	0	0	0	0	0	0	0	0	2
	CDD50	11488	761	792	1037	1124	1108	1034	1025	1008	984	990	862	763
	CDD65	6016	297	372	572	674	643	584	560	543	534	525	412	300
	CDH74	63327	2522	3714	6748	8725	7726	6667	5904	5404	5232	4836	3439	2410
	CDH80	28796	1036	1876	3784	5064	3911	2894	2222	1857	1875	1907	1435	935
Wind	WSAvg	1.8	1.0	1.5	2.3	2.7	2.4	2.7	2.6	2.1	1.4	1.0	1.0	1.1
Precipitation	PrecAvg	44.9	0.4	0.2	1.0	2.1	6.9	4.8	5.7	11.5	7.5	4.1	0.8	0.2
	PrecMax	61.3	2.8	0.7	4.2	4.0	13.9	11.2	11.5	17.8	11.1	8.8	3.0	0.7
	PrecMin	31.7	0.0	0.0	0.0	0.0	2.0	1.8	2.9	7.2	2.8	0.9	0.2	0.0
	PrecStd	8.6	0.8	0.3	1.2	1.5	3.3	2.3	2.2	3.7	2.6	2.2	0.9	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.8	97.2	102.1	105.9	104.9	98.7	96.5	93.5	94.6	95.3	94.7	93.2
		MCWB	73.0	73.4	74.4	77.2	77.3	78.7	79.7	80.8	80.9	80.0	76.8	75.0
	2%	DB	90.4	95.1	99.7	102.6	100.3	95.4	93.3	91.7	92.2	92.5	92.2	90.0
		MCWB	71.9	73.0	74.6	77.7	78.7	79.4	79.7	80.1	80.2	79.3	76.3	73.8
	5%	DB	88.5	93.1	97.8	100.2	97.1	93.2	91.3	90.0	90.5	91.0	90.0	87.9
		MCWB	71.3	72.8	74.7	77.7	79.0	79.3	79.3	79.4	79.7	78.7	75.6	73.0
	10%	DB	86.3	91.0	95.4	97.7	94.5	91.1	89.3	88.2	88.8	89.3	88.0	85.7
		MCWB	70.6	72.5	74.6	77.6	79.0	79.0	78.8	79.0	79.3	78.2	74.9	71.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.9	78.1	80.5	82.3	83.1	82.4	82.2	82.5	82.8	82.5	79.9	78.1
		MCDB	88.1	92.3	96.2	97.5	93.7	93.1	91.0	90.4	90.9	91.4	89.2	88.4
	2%	WB	74.9	76.4	78.7	80.8	81.7	81.0	80.9	81.2	81.5	80.8	78.5	76.0
		MCDB	85.0	89.9	93.7	96.3	92.8	90.6	89.6	88.9	89.2	89.7	88.0	85.8
	5%	WB	73.4	75.2	77.4	79.8	80.8	80.3	80.0	80.2	80.6	79.9	77.4	74.5
		MCDB	83.7	88.4	92.2	94.9	91.9	89.4	88.2	87.5	88.3	88.2	86.2	83.9
	10%	WB	72.0	73.9	76.2	78.8	79.9	79.7	79.2	79.5	79.8	79.1	76.4	73.2
		MCDB	82.2	86.5	90.4	93.5	90.4	88.2	86.8	86.1	86.7	86.7	84.2	82.2
Mean Daily Temperature Range	5% DB	MDBR	25.8	27.2	25.2	21.8	16.8	14.0	12.5	12.1	13.3	15.6	19.5	23.1
		MCDBR	28.1	29.6	28.2	24.1	21.0	17.0	15.7	14.6	15.6	17.8	21.5	24.3
		MCWBR	11.0	10.6	8.8	7.0	5.9	5.2	5.3	5.5	6.0	6.7	7.7	9.4
	5% WB	MCDBR	23.4	25.5	24.8	21.8	17.7	14.9	14.1	13.6	14.6	16.3	18.6	21.6
		MCWBR	9.6	9.8	8.8	7.4	6.3	5.5	5.5	5.6	6.2	6.8	7.0	8.7
Clear-Sky Solar Irradiance	taub	0.455	0.553	0.674	0.645	0.500	0.446	0.437	0.451	0.486	0.530	0.444	0.444	
	taud	2.156	1.908	1.656	1.771	2.191	2.368	2.388	2.330	2.218	2.051	2.264	2.213	
	Ebn at Noon	260	240	217	225	257	268	271	269	260	244	263	260	
	Edh at Noon	45	61	81	73	47	39	38	41	46	52	41	42	
All-Sky Solar Radiation	RadAvg	1449	1642	1722	1880	1722	1543	1430	1407	1465	1474	1499	1394	
	RadStd	77	67	102	109	148	117	104	88	72	80	81	68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+324	+328
Regional (20 neighbors)		+0.47	N/A	N/A	N/A	N/A	+0.22	N/A	N/A	+225	+227

Nomenclature: See separate page